UNIT 5 · STANDARD 6.GR.1

Determine the Area of Triangles

Lesson 5-2

Name: _____ **Date:** _____ **Class:** _____

SKY HARBOR MISSION

The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a triangle using the formula A = ½ × base × height.

Language Objective: I can explain my steps using the words base, height, area, and perpendicular.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Base · **Height** · **Area** · **Perpendicular** · **Composite figure** · **Formula** · **Parallel** · **Substitute**

Level 3: try the blanks from memory first, then check the bank.

1. The side of a triangle you measure the height from at a 90° angle is the _____ .

2. The perpendicular distance from the base to the opposite vertex is the _____ .

3. The amount of flat space inside a triangle is its _____ .

4. Two lines that meet at a 90° angle are _____ .

5. A shape made of two or more basic shapes combined is a _____ .

6. A math rule written with symbols, like A = ½ × b × h, is a _____ .

7. Two sides that stay the same distance apart and never meet are _____ .

8. Putting known numbers into a formula is called _____ .

Write About the Math

How many cans of paint does the homeowner need?
¿Cuántas latas de pintura necesita el dueño de la casa?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Base** — *Base*
- ☐ **Height** — *Altura*
- ☐ **Area** — *Área*

- [ ] **Perpendicular**
  Perpendicular
- [ ] **Composite figure**
  Figura compuesta

Model: *The height is the perpendicular distance from the base to the top corner.* — Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The wall's area is ___ sq ft because A = 1/2 × ___ × ___. She needs ___ cans because ___ ÷ 50 = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.